STELLAR NUCLEOSYNTHESIS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Stellar Nucleosynthesis

Question 1

10.0 points

Examine the proton-proton (p-p) chain reaction depicted in the diagram below for main sequence stars. Identify and explain the sequence of nuclear reactions occurring in the p-p chain, including the role of neutrinos and gamma rays in energy transport within the star.

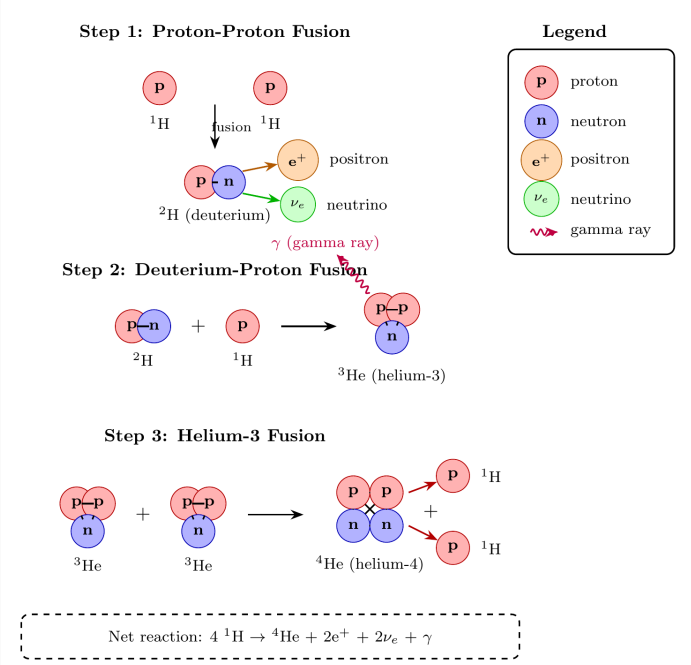

Fig. 1. A diagram of the proton-proton (p-p) chain reaction in main sequence stars. The diagram should show the sequence of nuclear reactions starting from two protons (1H) fusing to form deuterium (2H), followed by the production of helium-3 (3He), and finally the fusion of two helium-3 nuclei to form helium-4 (4He). Include the emission of positrons, neutrinos, and gamma rays at each step. Label each particle and reaction clearly, and indicate the role of neutrinos and gamma rays in energy transport within the star. IMPORTANT correction needed: The diagram contains labels ("neutrinos escape", "gamma rays heat plasma") that directly state the roles of these particles in energy transport, which is part of the explanation the student is required to provide.. Specifically: The labels "neutrinos escape" and "gamma rays heat plasma". Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 2

10.0 points

Examine the CNO cycle reaction network in the diagram below for massive stars. Describe how this cycle contributes to the stellar nucleosynthesis process and compare its efficiency to the proton-proton chain in terms of energy production.

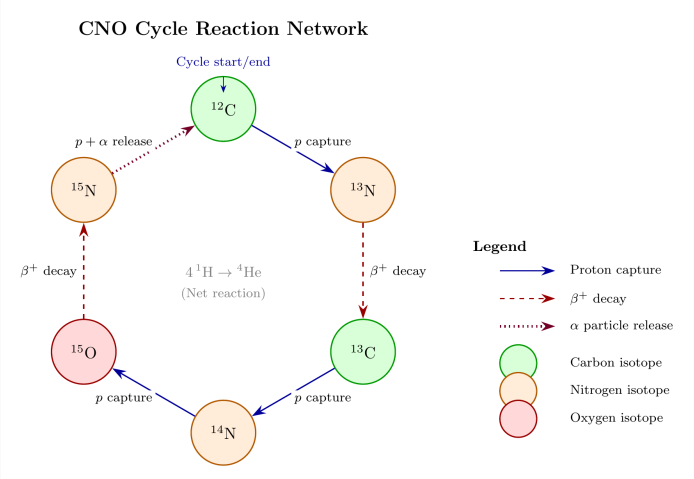

Fig. 2. Diagram of the CNO cycle reaction network for massive stars. Include the main nuclei: 12C, 13N, 13C, 14N, 15O, 15N. Show the sequence of reactions: proton captures and beta decays. Indicate the starting and ending points of the cycle. Highlight energy production steps with arrows and labels.

Question 3

10.0 points

Interpret the Hertzsprung-Russell (H-R) diagram with evolutionary tracks shown in the diagram below. Explain how this diagram represents the life cycles of different mass stars and identify the key stages for a one-solar-mass star.

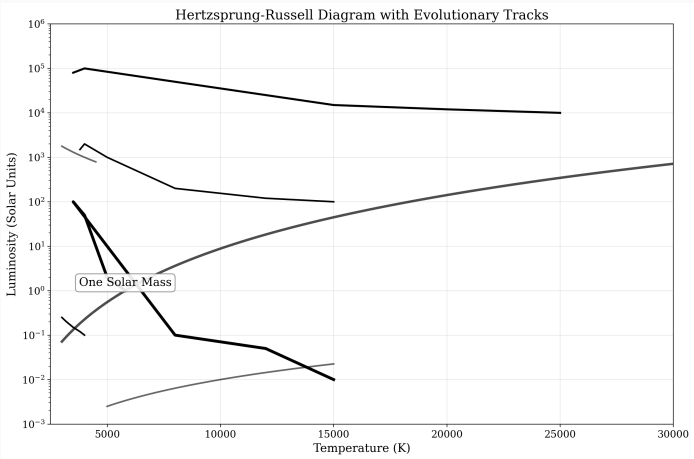

Fig. 3. Hertzsprung-Russell (H-R) diagram showing evolutionary tracks for stars of different masses. The diagram includes axes labeled for luminosity and temperature, with tracks highlighting the path of a one-solar-mass star. Key stages such as the main sequence, red giant branch, and white dwarf stage are marked.

Question 4 10.0 points

Analyze the binding energy per nucleon curve in the diagram below, focusing on the iron peak. Discuss how this curve illustrates the energy production limits in stellar fusion and why iron represents an endpoint for nucleosynthesis in stars.

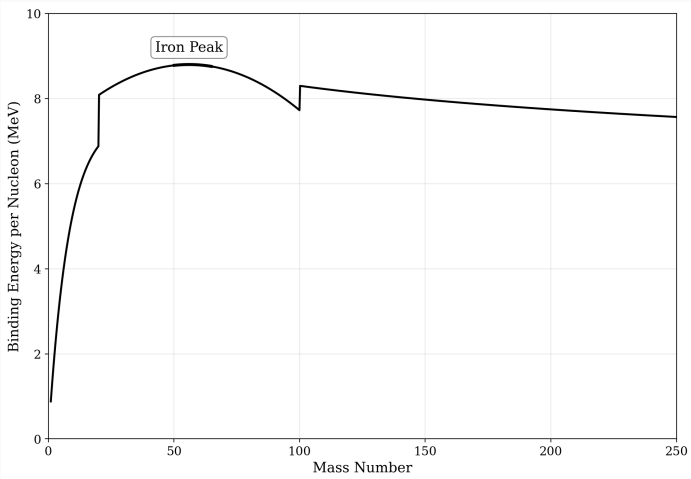

Fig. 4. Binding energy per nucleon curve showing the stability of nuclei. The x-axis represents the mass number (A), and the y-axis represents the binding energy per nucleon (MeV). The curve peaks around iron (Fe), illustrating the maximum binding energy per nucleon, which signifies the endpoint for nucleosynthesis in stars. The diagram highlights the iron peak and the general trend of increasing binding energy up to iron and decreasing beyond it.

Question 5 10.0 points

Examine the onion-shell model diagram of a pre-supernova massive star's internal structure shown below. Describe the different shells and their associated nuclear processes, explaining how this structure leads to a supernova event.

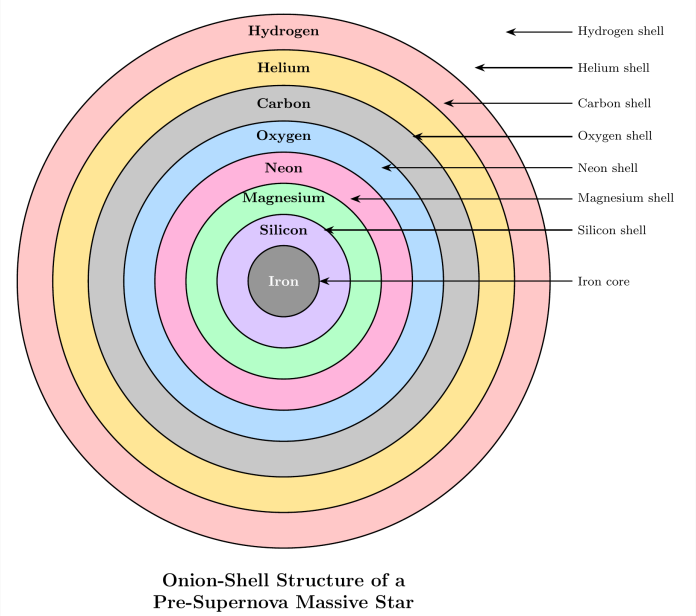

Fig. 5. An onion-shell model diagram of a pre-supernova massive star. The diagram should show concentric layers representing different shells. From outermost to innermost, the shells are: hydrogen, helium, carbon, oxygen, neon, magnesium, silicon, and the iron core. Each shell must be labeled with both its element name and its associated nuclear process: the hydrogen shell with "hydrogen burning (H → He)", the helium shell with "helium burning (He → C)", the carbon shell with "carbon burning (C → O)", the oxygen shell with "oxygen burning (O → Ne)", the neon shell with "neon burning (Ne → Mg)", the magnesium shell with "magnesium burning (Mg → Si)", and the silicon shell with "silicon burning (Si → Fe)". The innermost core should be labeled "Iron Core".